

GTP release-selective agonists prolong opioid analgesic efficacy

<https://doi.org/10.1038/s41586-025-09880-5>

Received: 2 April 2025

Accepted: 6 November 2025

Published online: 17 December 2025

Open access

 Check for updates

Edward L. Stahl¹, Matthew A. Swanson^{1,2}, Vuong Q. Dang³, Michael D. Cameron³, Nicole M. Kennedy³, Thomas D. Bannister³ & Laura M. Bohn^{1,2}

G-protein-coupled receptors act as guanine nucleotide exchange factors (GEFs) and facilitate the activation of heterotrimeric G proteins by exchanging GDP for GTP¹. This exchange function is not unidirectional². Here we demonstrate that an agonist can show selective affinity for an active state that prefers the release of GTP. Specifically, for the mu opioid receptor, we show that several agonists have state-selective affinities for promoting GTP release versus GTP binding. We identify two agonists that show a marked preference for promoting release. In mice, marginally efficacious doses of the release-preferring agonist enhance and prolong the antinociceptive effects of morphine and fentanyl without enhancing the respiratory and cardiac effects of fentanyl. Although these observations are limited to simple measures of thermal nociception, they may point to a way to bifurcate physiological responses to such agonists. We propose that the active-state selectivity of an agonist may determine the preferred direction of the receptor GEF function, which may affect the kinetics and selectivity of the engagement of the receptor with downstream effectors; this may ultimately present a means to disentangle multifaceted drug-induced physiological responses.

Heterotrimeric G proteins transduce information to intracellular partners by modulating GTP binding and hydrolysis³. Through their interaction with G-protein-coupled receptors (GPCRs) and effectors, G proteins provide the transducer function that is necessary for the conveyance of extracellular information^{4,5}. Heterotrimeric G proteins consist of an α subunit bound to a β and γ subunit dimer; they remain a trimer while the α subunit is bound to GDP⁶. Receptors provide the transmembrane conduit for a signal between the extracellular agonist and the intracellular G-protein transducer⁷. Specifically, GPCRs undergo a conformational change that acts to catalyse a reaction between the receptor and the $G\alpha$ protein^{8–10}. This interaction shifts the affinity for $G\alpha$ binding to GDP to conditions that favour GDP release and GTP binding³. Thus, the receptor acts as a GEF and this reaction is considered to be primarily unidirectional¹¹ (Fig. 1a). However, there have been observations that the GTP loading function of the receptor is reversible—that is, the receptor may facilitate the release of GTP from $G\alpha$. Early examples of this reversible interaction used nonhydrolysable forms of GTP such as GTP γ S^{3,12}, wherein the dissociation of radiolabelled GTP γ S could be observed upon agonist binding to the receptor. One such study examined the kinetics of the release of ³⁵S-GTP γ S in cells expressing the mu opioid receptor (MOR) and found that the rate of release of nucleotide was increased as a function of a single saturating concentration of agonist and that partial and full agonists maintained their rank order efficacy in both exchange reactions¹³ (³⁵S-GTP γ S binding and ³⁵S-GTP γ S release).

Here we investigate the significance of the release mechanism as a function of agonist concentration and how it can influence drug

responsiveness in vivo. In a linked Article¹⁴, we present extensive pharmacological and biochemical characterizations of the release reaction, which we summarize in the three-state coupling model (Fig. 1a). In the study, we show that the GTP-release function of the receptor adheres to the pharmacological principles that pertain to the GTP-binding function of the receptor. In summary, the release function is dependent on agonist concentration and can be reversed by antagonists, and competitive interactions are preserved between orthosteric agonists and antagonists. Moreover, we demonstrate that the effect is due to activation of the receptor population and not merely a function of receptor occupancy. We provide experimental evidence and a functional state model that establishes that the efficacy and potency of an agonist to promote GTP release can differ from its efficacy and potency to induce GTP binding. Therefore, an agonist may demonstrate selectivity for affecting the equilibrium of the functional active state of the G protein and can show a preference for one state over the other. We also provide evidence that an agonist may have a different rank order potency and efficacy for the two states of the exchange function of a GPCR.

In the present study we show that agonists can induce both GTP binding and GTP release from $G\alpha$ in a concentration-dependent manner, and that this can be observed for several different GPCRs (Fig. 1). GTP binding is assessed using a conventional method that entails incubating isolated cell membranes in the presence of ³⁵S-GTP γ S and increasing concentrations of agonists^{15,16}. To observe GTP release, we use a ‘pulse-chase’ paradigm, which entails first loading the membrane preparations with ³⁵S-GTP γ S. Since many GPCRs are negatively regulated

¹Department of Molecular Pharmacology and Physiology, Morsani College of Medicine, University of South Florida, Tampa, FL, USA. ²The Skaggs Graduate School of Chemical and Biological Sciences at Scripps Research, La Jolla, CA, USA. ³Department of Molecular Medicine, The Herbert Wertheim UF Scripps Institute for Biomedical Innovation and Technology, Jupiter, FL, USA.

[✉]e-mail: EStahl@usf.edu; LBohn@usf.edu

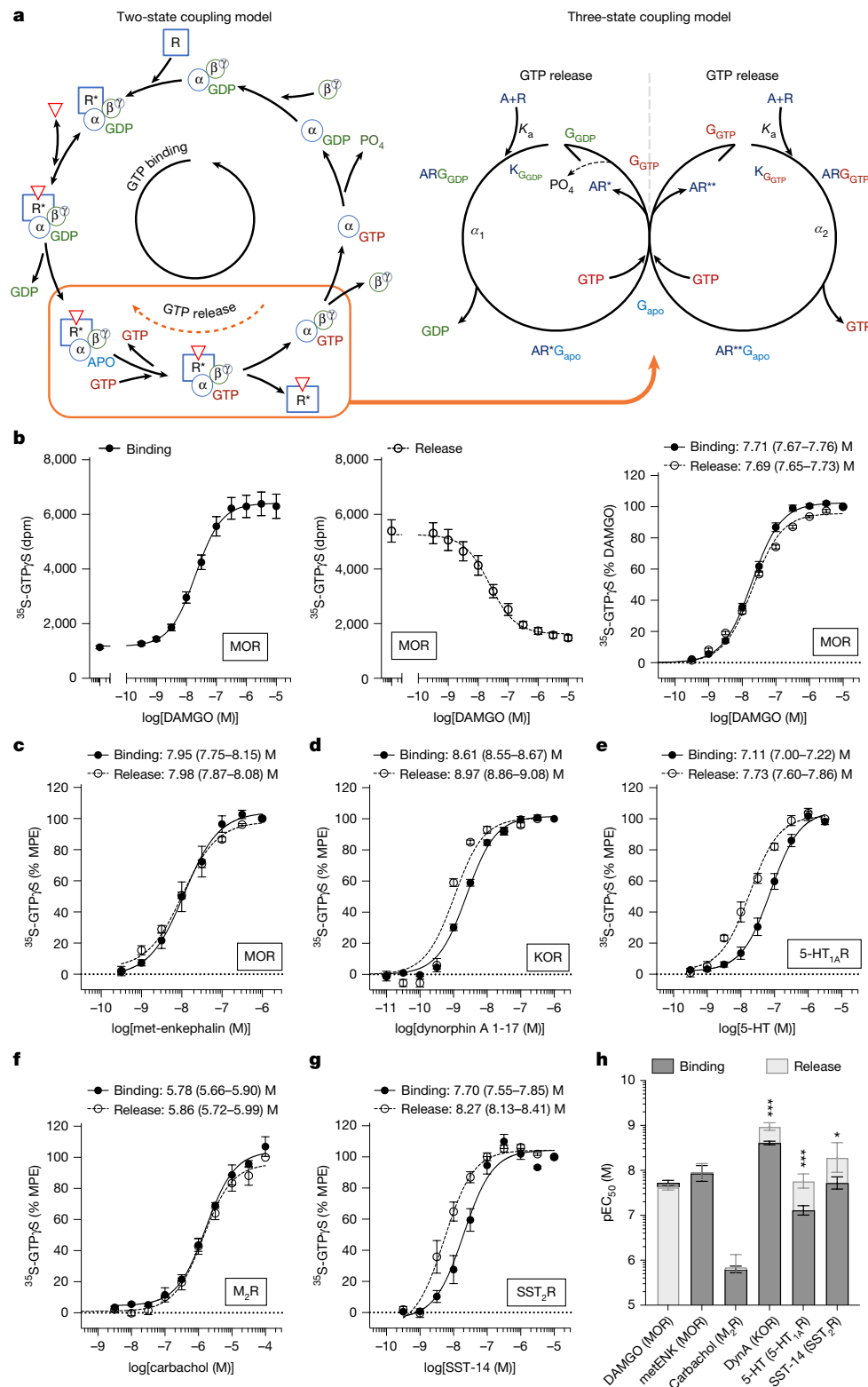


Fig. 1 | GPCRs induce both GTP binding and GTP release from G α proteins and the process is agonist-mediated. **a**, Schematic of the proposed model, showing the conventional pathway of GDP-to-GTP exchange (left) and the expanded model to allow for both GTP and GDP release (right) as detailed in the linked Article¹⁴. A, agonist; R, receptor (asterisks indicate different active states); K_a , affinity constant; G protein; G_{apo}, unbound G protein (blue); G_{GDP}, GDP-bound G protein (green); G_{GTP}, GTP-bound G protein (red); α_1 and α_2 , active state affinities. **b–g**, DAMGO-stimulated binding and release in CHO-MOR cells presented as raw data (in disintegrations per minute (dpm)) and the normalization to baseline and maximum response. **c–g**, Normalized binding

and release with indicated agonists in CHO-K1 cells expressing MOR (**c**; $n = 3$ binding, 3 release), KOR (**d**; $n = 3$ binding, 5 release), 5-HT_{1A}R (**e**; $n = 8$ binding, 7 release), M₂R (**f**; $n = 3$ binding, 3 release) and SST₂R (**g**; $n = 4$ binding, 4 release). The raw data are presented in Extended Data Fig. 1. MPE, maximum possible effect. **b–g**, Data are mean $\pm$ s.e.m. and potency is presented as mean with 95% confidence interval. **h**, Comparison of potency in ³⁵S-GTP γ S binding versus release by unpaired, two-tailed *t*-test for each receptor comparing the individual potency (pEC₅₀, where EC₅₀ is half-maximal effective concentration) values measured per experiment. Data are mean with 95% confidence interval. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001; NS, not significant (*P* > 0.05).

by sodium ions, removal of sodium allows for the constitutive activation of all sensitive receptors and the subsequent loading of ^{35}S -GTP γ S binding to G proteins. After the pulse, the chase entails dilution of the membranes and inclusion of an excess of unlabelled (cold) GTP γ S in the presence of sodium (see Methods).

The two reactions are compared in Fig. 1b using membranes prepared from cells overexpressing mouse MOR. The data are presented as radioactivity counts for both the binding and the release assay; to facilitate comparison of the potencies, the data are also normalized to the baseline (0%) and the highest concentration used in each response (100%), and the curve is inverted for the release function. For the MOR, the potency of DAMGO ([D-Ala₂, N-MePhe₄, Gly-ol]-enkephalin), an enkephalin analogue, is conserved for both assays and the same is true for met-enkephalin (Fig. 1c). The exchange effect can also be observed for dynorphin A (1–17) at the kappa opioid receptor (KOR), serotonin (5-HT) at the serotonin 1A receptor (5-HT_{1A}R), carbachol at the muscarinic 2 receptor (M₂R) and somatostatin-14 (SST-14) at the somatostatin 2 receptor (SST₂R) (Fig. 1d–g). Notably, agonist potencies at MOR and M₂R are conserved in the two states, whereas dynorphin, serotonin and somatostatin are significantly more potent at their cognate receptors for promoting the release of GTP (Fig. 1h; individual curves are shown in Extended Data Fig. 1).

There are multiple clinically relevant opioid agonists that span a broad range of pharmacological characteristics (including partial agonists and biased agonists); therefore, we used these tool compounds to determine whether the release function and binding function could be dissociated at the MOR. In addition, we tested two new compounds, which were selected on the basis of their scaffold variation from biased MOR agonists introduced by our laboratory (the SR series—for example, SR-17018) and for their characteristics as full agonists that are less potent than morphine in cellular assays. The latter consideration was based on a desire to not introduce more potent opioid agonists to the scientific literature. For each drug, DAMGO was assayed in parallel to serve as a reference, since DAMGO maintains the same potency in both responses and serves to define the maximum efficacy in both assays in this cell line. Not unexpectedly, several agonists perform similarly to DAMGO, preserving the potency in both responses; however, some agonists show a differential preference for potency (Fig. 2a) and/or efficacy (Fig. 2b) for one state over the other (see Extended Data Fig. 2 for curves and Extended Data Table 1 for parameters). Since the two effects were measured in parallel with DAMGO, we also determined the difference in transduction efficiencies ($\Delta\Delta\log R$; Fig. 2c and Extended Data Table 1) for each agonist in the release assay and the binding assay. This representation permits normalization between responses to directly compare agonist activity¹⁷. The two new agonists show significant gains in the release function, having nearly a hundred-fold gain in selectivity for the release active state, as measured by the difference in transduction efficiencies; we have named these compounds muzepan1 and muzepan2 (Fig. 2d), as they are mu opioid receptor-acting compounds containing an 'azepane' ring.

Several of the agonists that show a state preference have previously been identified as biased agonists that prefer G-protein signalling over β -arrestin2 recruitment (for example, oliceridine¹⁸, PZM21¹⁹, herkinorin²⁰, buprenorphine^{21–23} and SR-17018²⁴). When tested in the cellular assays expressing the human MOR that were used to evaluate the biased agonism of SR-17018²⁴, both muzepan1 and muzepan2 show no preference between GTP γ S binding and β -arrestin2 recruitment (Extended Data Fig. 3a and Extended Data Table 2). Moreover, the exchange selectivity for GTP release over binding is maintained at the human receptor (Extended Data Fig. 3b, Extended Data Table 3). Therefore, whereas many of the compounds that showed selectivity for release over binding also show preference for G-protein binding over β -arrestin2 recruitment, the correlation is not absolute.

To demonstrate the physiological significance of agonist-induced GTP release, the experiment was repeated in mouse spinal cord

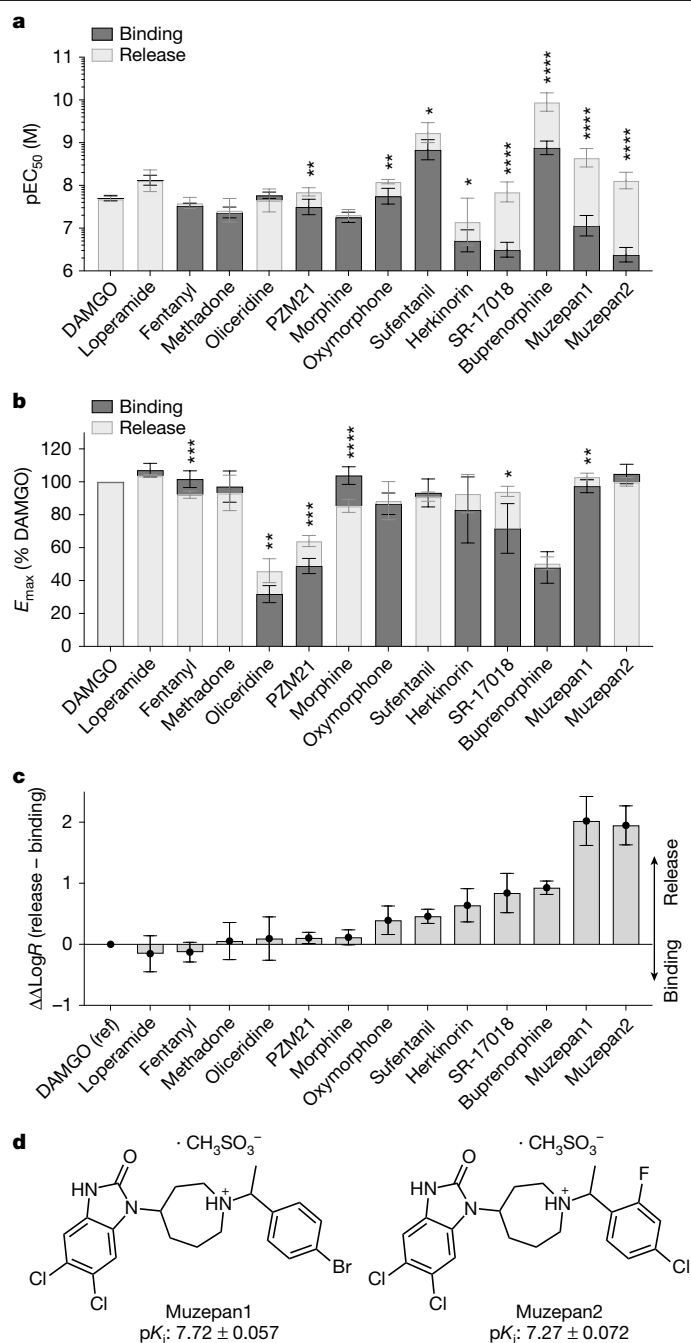


Fig. 2 | Opioid agonists exhibit differential preferences for GTP binding and release in CHO-MOR cell membranes. **a–c**, Comparisons of the mean of the individual ^{35}S -GTP γ S binding and release: potencies (**a**), maximum efficacies (E_{max} normalized to DAMGO (100% versus baseline (0%)) (**b**), and difference in transduction efficiencies ($\Delta\Delta\log R$, relative to DAMGO) (**c**). Data are mean with 95% confidence interval. Unpaired t -test was used for comparing binding and release parameters for each compound. Extended Data Fig. 2 shows concentration–response curves and Extended Data Table 1 presents parameters and number of individual replicates ($n \geq 3$). **d**, Chemical structures of muzepan1 and muzepan2 with binding affinities (pK_i with s.e.m., $n = 6$; K_i is the inhibition constant) determined from competition binding assays with ^3H -naloxone.

membranes. In the binding experiment, DAMGO promotes only a 40% stimulation in the native tissue (1.4 ± 0.01 -fold; $P < 0.001$, paired t -test versus baseline; Extended Data Fig. 4a). We determined that the sodium-free conditions lead to very high levels of GTP γ S binding, making it difficult to see an effect of DAMGO on release. This is not unexpected, as there are relatively low levels of MOR in the system,

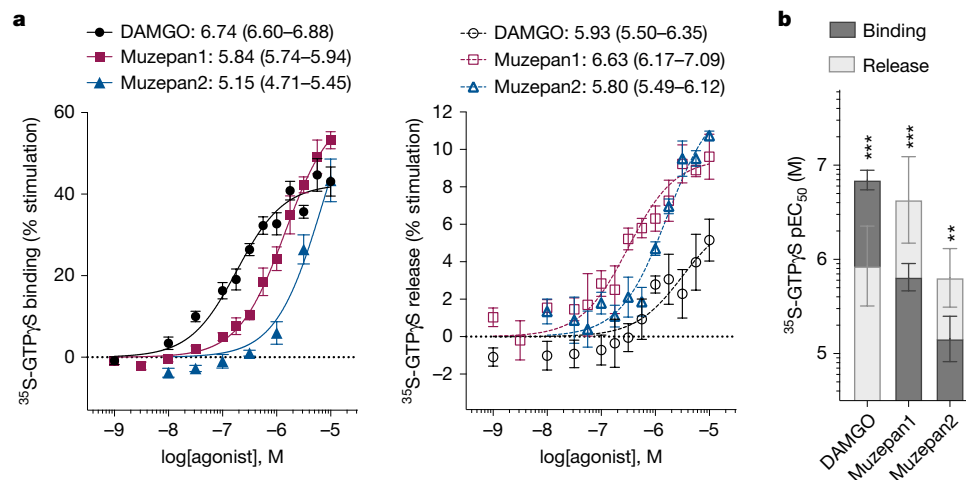


Fig. 3 | Bidirectional GTP exchange in mouse spinal cord membranes.

a, ^{35}S -GTP γ S binding and release induced by indicated agonists. Data points show mean $\pm$ s.e.m.; potencies are presented as pEC₅₀ with 95% confidence interval. Binding: DAMGO: $n = 8$; muzepan1: $n = 12$; muzepan2: $n = 4$.

Release: DAMGO: $n = 8$; muzepan1: $n = 9$; muzepan2: $n = 4$. **b**, Comparison of potencies between the binding and release exchange function for each agonist shown in **a**. Data are mean with 95% confidence interval. Unpaired, two-tailed, t -test of the mean of individual experiments.

as reflected by the approximately 40% stimulation in the binding studies (Extended Data Fig. 4a). Therefore, to isolate the MOR-accessible G-protein pool, we used DAMGO in the presence of sodium and ^{35}S -GTP γ S in the pulse phase and diluted 100-fold as part of the chase (Methods and ref. 14). We demonstrate that this is feasible in the CHO-mMOR cell line, where the potencies of DAMGO are similar to those in the sodium-free loading conditions, although the potency for the release function is slightly decreased (19 nM binding versus 43 nM release; $P < 0.05$, t -test; Extended Data Fig. 4b). In mouse spinal cord membranes, when 100 nM DAMGO is included in the pretreatment period, there is an increase of about 10% in ^{35}S -GTP γ S loading ($P < 0.001$; paired t -test), but no change is evident in the spinal cord of MOR-knockout mice (Extended Data Fig. 4c). We therefore took this modest stimulation as representative of MOR-mediated GTP γ S loading in the mouse spinal cord.

In mouse spinal cord membranes, DAMGO is more potent in stimulating ^{35}S -GTP γ S binding than muzepan1 and muzepan2, whereas all agonists are full agonists (Fig. 3a). In the release paradigm, DAMGO loses potency, whereas muzepan1 and muzepan2 gain potency (Fig. 3b). Notably, the efficacy obtained by muzepan1 and muzepan2 reach the 10% maximal effect anticipated in the pulse loading (Extended Data Fig. 4c). This is in contrast to DAMGO, which does not reach this plateau, suggesting that in spinal cord, the enkephalin-like agonist may be selective against release. No significant effects were observed in spinal cord membranes from MOR-knockout mice (Extended Data Fig. 4d); therefore, the effects are likely to be due to MOR activation.

In mice, muzepan1 and muzepan2 are brain penetrant as they can be detected in brain 1 h following intraperitoneal injection of 3 mg kg⁻¹ of compound (muzepan1: 463 $\pm$ 83 nM, muzepan2: 493 $\pm$ 187 nM, $n = 3$; Extended Data Fig. 5a). We therefore tested the compounds for antinociceptive efficacy in the hot plate and warm water tail immersion assays and compared them with morphine (Fig. 4a). The potencies determined by the effect at 1 hour correlate with their rank order potency in GTP γ S binding (as well as release), although we note that at higher doses, muzepan1 remains nearly maximally efficacious for the 4 h duration of the test (Fig. 4b). Notably, none of the agonists have effects in MOR-knockout mice (Fig. 4b,c; for morphine see ref. 24).

Since muzepan1 and muzepan2 promote GTP release more potently than GTP binding, we considered whether a sub-efficacious dose would alter responsiveness to conventional opioid analgesics. When co-administered with 12 mg kg⁻¹ morphine, a low dose of muzepan1

(3 mg kg⁻¹) enhances and prolongs morphine-induced antinociception, and exceeds the calculated additive individual effects of each drug in both nociceptive tests (Fig. 5a). Furthermore, the inclusion of muzepan1 at varying doses of morphine produces the same enhancement, significantly improving the potency (median effective dose (ED₅₀)) of morphine by about two-fold ($P < 0.0001$; Fig. 5b and Extended Data Fig. 6) in both assays. Muzepan2 (12 mg kg⁻¹), which alone produces very little antinociception, also enhances and prolongs the response of morphine in a manner that is greater than additive (Fig. 5c).

To assure that the compounds are not indirectly enhancing morphine effects by competing with morphine metabolism, which occurs via glucuronidation²⁵, we also used fentanyl, which is metabolized by CYP3A4²⁶. Notably, neither compound competes with CYP3A4 as determined by in vitro competition studies (muzepan1: 11%, muzepan2: <10% inhibition at 10 μM ; Extended Data Fig. 5b). Whereas only male mice were tested with morphine owing to sex-dependent differences in morphine metabolism and sensitivity²⁷, both male and female mice were tested with fentanyl, since the responses that we measured are comparable between the sexes^{24,28}. When combined, muzepan1 dramatically prolongs the efficacy of fentanyl over time in a manner that is greater than the predicted additive effects of both compounds (Fig. 5d; see Extended Data Table 4 for statistical analyses of all time course data in Fig. 5).

Notably, the effects of muzepans on opioid-mediated antinociception resemble those of adding a positive allosteric modulator to an MOR agonist. Therefore, we tested whether muzepan1 could act allosterically by measuring its concentration-dependent effect on fentanyl-induced GTP γ S binding in spinal cord membranes. Since we do not observe a leftward shift in fentanyl potency, we cannot conclude that muzepan1 behaves as a positive allosteric modulator at MOR (Extended Data Fig. 7).

Whereas it is desirable to enhance the antinociceptive effects of analgesic opioids, it is not desirable to enhance the respiratory suppression and bradycardia associated with these agonists. Therefore, we tested the effect of muzepan1 alone and in combination with fentanyl in mouse pulse oximetry and heart rate monitoring studies (Fig. 6). Alone, muzepan1 produces respiratory suppression and bradycardia at 24 mg kg⁻¹ and 48 mg kg⁻¹ but not at 3 mg kg⁻¹; by contrast, fentanyl at lower doses (0.3 mg kg⁻¹ and 2 mg kg⁻¹) produces marked decreases in arterial oxygen saturation and heart rate (two-way repeated measures ANOVA, see Extended Data Table 5 for all comparisons). The response to the combination of 3 mg kg⁻¹ muzepan1

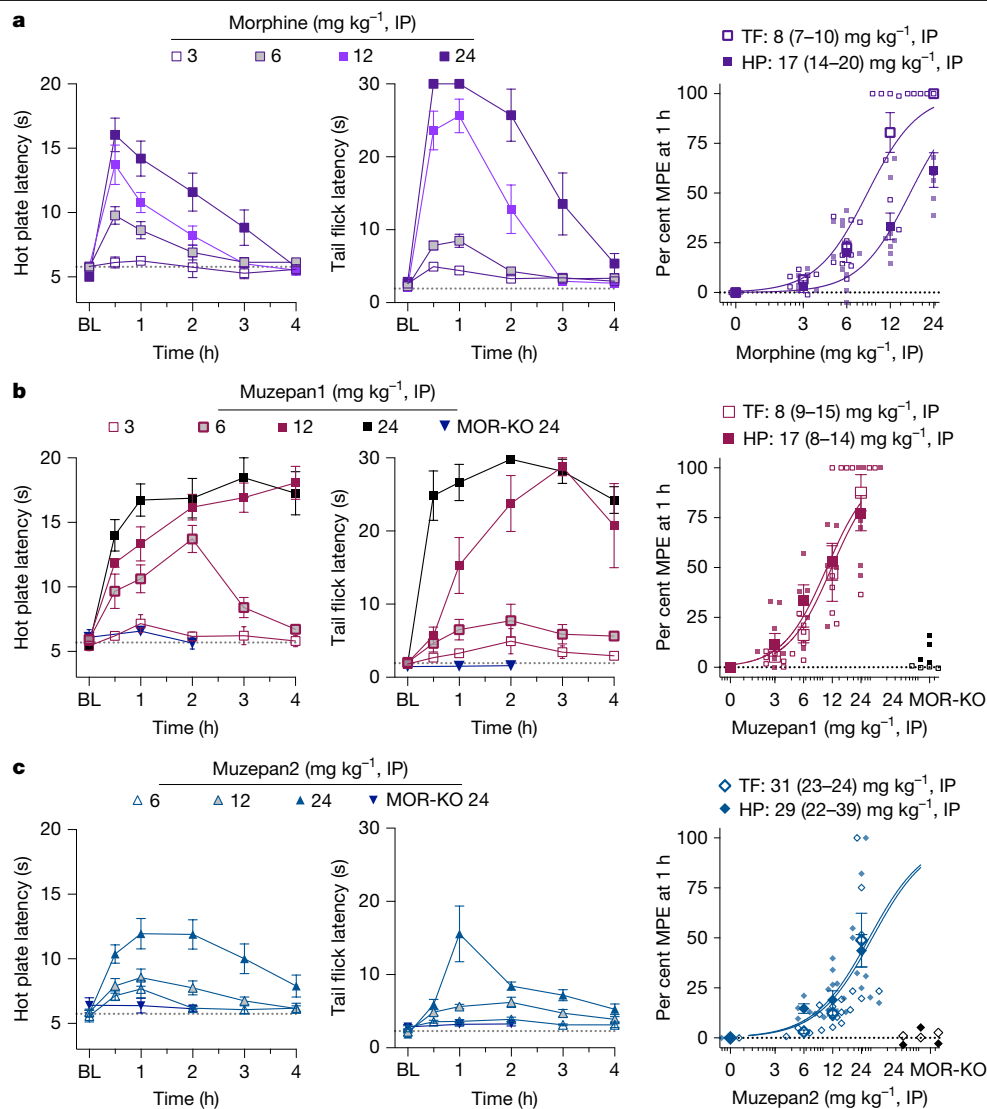


Fig. 4 | Potency in mouse hot plate and warm water tail immersion nociceptive assays. **a–c**, Morphine (**a**), Muzepan1 (**b**) and Muzepan2 (**c**) were tested at the indicated doses (in mg kg^{-1} , intraperitoneal injection (IP)) in the hot plate (left) and tail flick (middle) assays in wild-type and MOR-knockout (MOR-KO) mice. Right, potency was determined by comparing the response at 1 h based on the percentage of MPE calculated from the baseline (BL) and a cut-off time of 20 s for the hot plate and 30 s for the tail flick. Potency curves

are extended to convey 0% (determined by baseline) and 100% (determined by maximum cut-off time used to estimate the ED_{50}); potencies are presented with 95% confidence interval in the legends. **a**, Morphine: $n = 6$ (3 mg kg^{-1} and 24 mg kg^{-1}), $n = 10$ (6 mg kg^{-1} and 12 mg kg^{-1}). **b**, Muzepan1: $n = 8$ (3 mg kg^{-1}), $n = 5$ (6 mg kg^{-1} and 12 mg kg^{-1}), $n = 7$ (24 mg kg^{-1}); MOR-KO: $n = 5$. **c**, Muzepan2: $n = 4$ (6 mg kg^{-1}), $n = 10$ (12 mg kg^{-1}), $n = 6$ (24 mg kg^{-1}); MOR-KO: $n = 3$. HP, hot plate assay; TF, tail flick assay.

with 0.3 mg kg^{-1} or 2 mg kg^{-1} fentanyl does not differ from the effect produced by fentanyl alone in either assay. Furthermore, increasing the dose of muzepan1 to 24 mg kg^{-1} , a dose that produced mild but significant effects alone (two-way repeated measures ANOVA; Fig. 5 and Extended Data Table 5), is also not additive with fentanyl. Therefore, inclusion of muzepan1 with fentanyl prolongs fentanyl-induced antinociception, whereas respiratory and cardiac effects are not enhanced.

Discussion

The state-dependent interaction between GPCRs and G proteins that we reveal here implies an ongoing cycle that would affect the availability of the G proteins and their ability to engage with different effectors. Moreover, steric hindrance produced by a constitutive G-protein cycle may prevent the receptor from interacting with other effectors, particularly with direct binders, such as β -arrestins. It is

attractive to speculate that persistence of the cycle could prevent the recruitment of β -arrestins. This is consistent with the observation that many of the agonists that produce state-dependent increases in potency and/or efficacy of GTP release, have been characterized as G-protein-signalling-biased agonists (that is, those that are biased against recruiting β -arrestins). However, here we present compounds that diverge from this correlation. Specifically, we focus on two full agonists that do not show bias against β -arrestin recruitment, and demonstrate that the release state selectivity cannot be the only contributor to G-protein signalling bias. Although the exceptions can be identified, it remains possible that changes in the G-protein cycling kinetics will affect GPCR signalling bias.

Our study demonstrates that all opioid agonists tested in the transfected cell system promote GTP release and that the potencies for both release and binding are similar for many agonists. In mouse spinal cord membranes, agonists are less potent in the binding assays compared with in the transfected cell system; this is probably due to

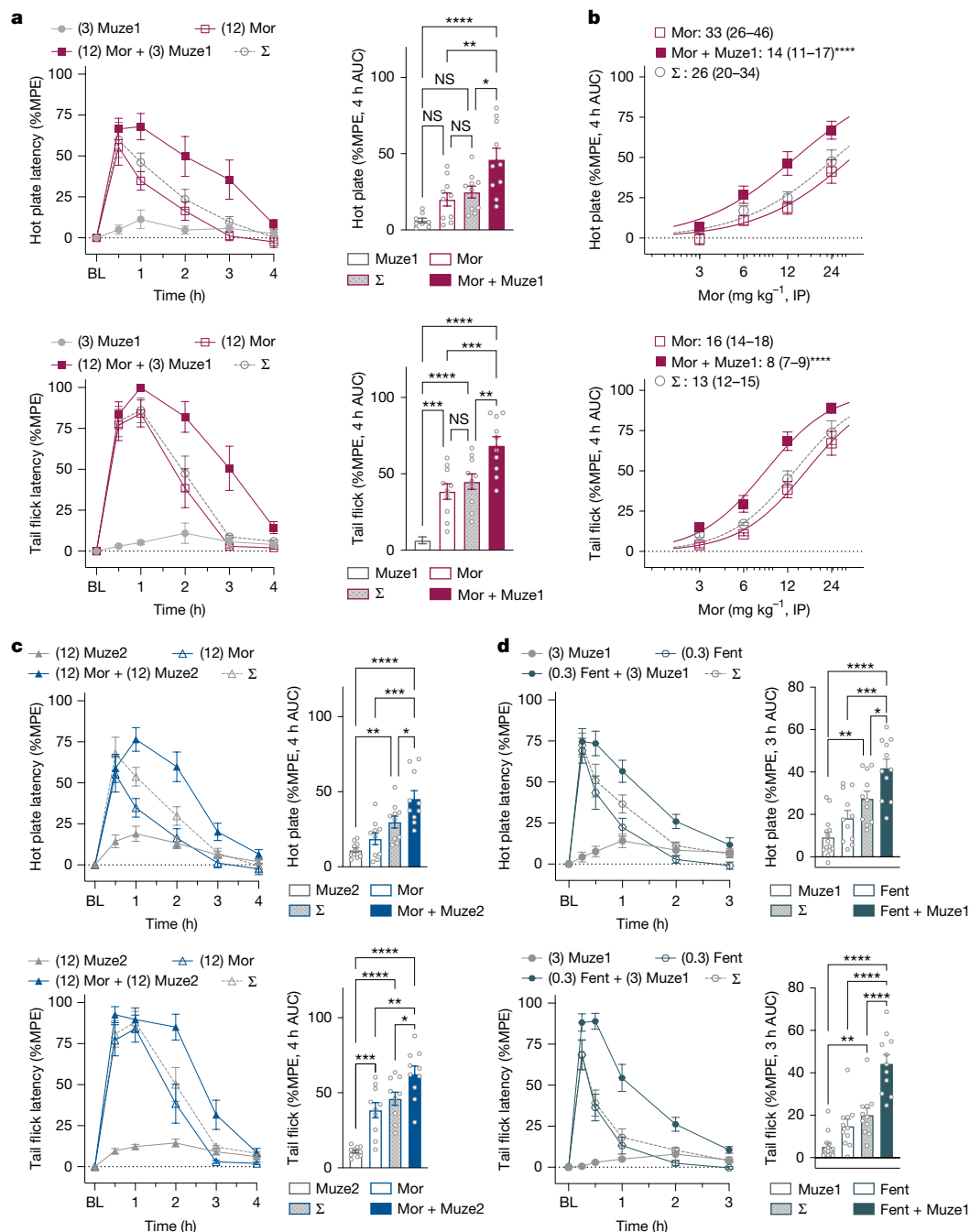


Fig. 5 | GTP-release-selective agonists enhance and prolong opioid-induced antinociception in mice. **a**, Hot plate (top) and tail flick (bottom) assays with muzepan1 (Muze1; 3 mg kg⁻¹, intraperitoneal) and morphine (Mor; 12 mg kg⁻¹, intraperitoneal) alone and combined. The calculated sum effect of both drugs (Σ) is shown for comparison. Right, mean ($\pm$ s.e.m.) area under the curve (AUC). One-way ANOVA. **b**, AUC for 4 h hot plate (top) and tail flick (bottom) assays following treatment with different doses of morphine (Extended Data Fig. 6). Legend indicates the ED₅₀ (with 95% confidence interval). **c**, Muzepan2

(Muze2; 12 mg kg⁻¹, intraperitoneal) also enhances morphine (12 mg kg⁻¹, intraperitoneal)-induced antinociception (analysis as in **a**). **d**, Muzepan1 (3 mg kg⁻¹, intraperitoneal) prolongs fentanyl (Fent; 0.3 mg kg⁻¹)-induced antinociception in male and female mice (analysis as in **a**). See Extended Data Table 4 for two-way repeated measures ANOVA analyses of time course data for **a–d** and the number and sex of mice in each assay ($n = 6–14$; individual mouse data are shown as symbols in the bar charts) as well as the results of post hoc ANOVA analyses for drug effect over time.

the low representation of MOR-expressing neurons in the gross dissection of the spinal cord. In the same preparations, DAMGO-induced release is significantly diminished, and this decrease may be amplified by the low number of receptors in the preparation. Regardless of the concentration of receptor, the release induced by muzepons remains efficacious, and their potency is significantly improved, suggesting that the robust effect on release is preserved even in this heterogeneous endogenous system.

In this system, we have utilized a nonhydrolysable form of GTP (GTP γ S) to demonstrate that the GEF function of the GPCR is reversible and that agonists that show a preference for the reversal can have effects at sub-efficacious doses in vivo. It is important to acknowledge that, in the endogenous setting, we must consider that the GTP bound to the G protein is not stabilized and is still subject to hydrolysis by the intrinsic GTPase activity of the G protein. Further, this can be acted upon by other endogenous GAPs such as regulator of G-protein signalling

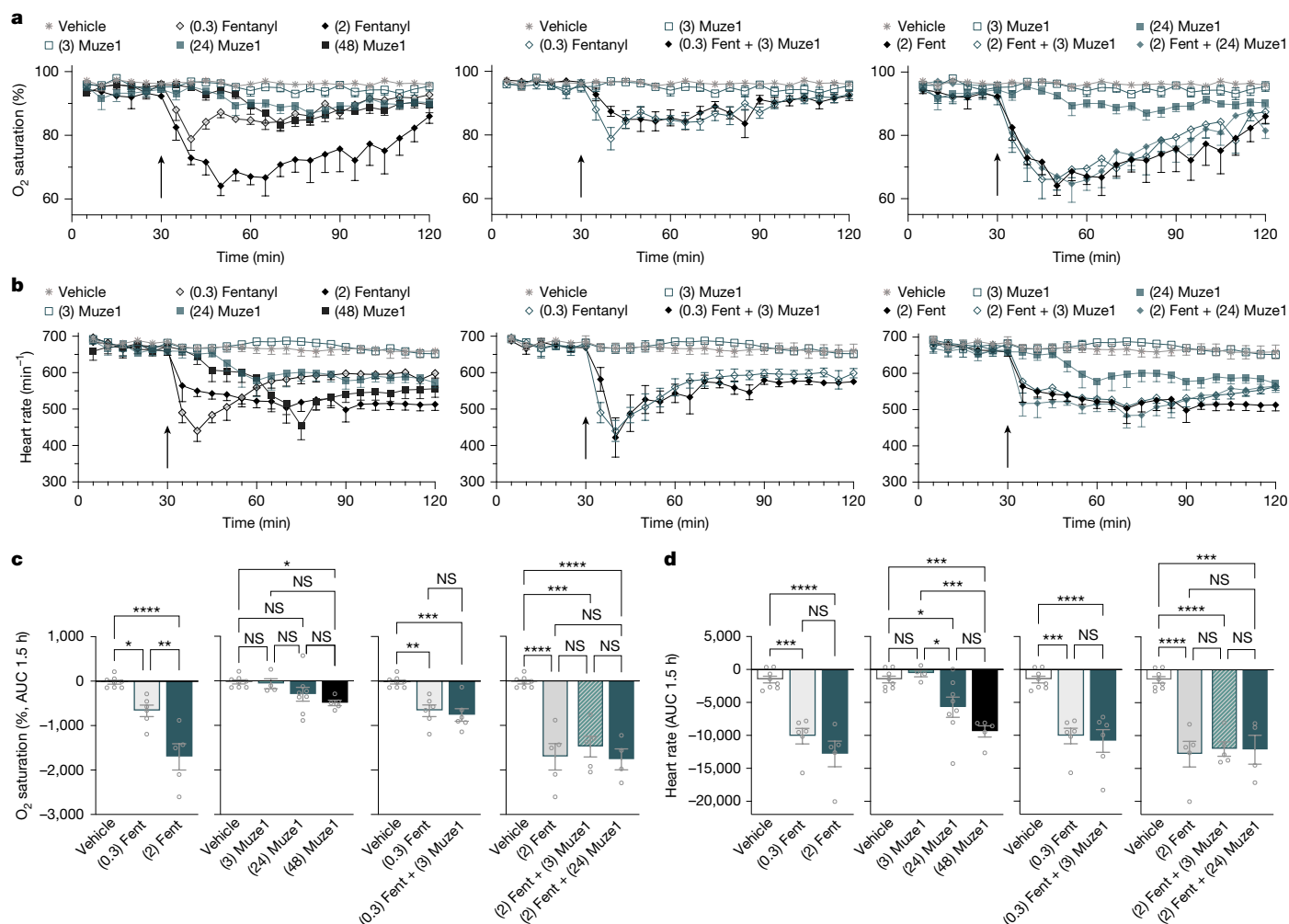


Fig. 6 | Muzepan1 does not enhance fentanyl-induced respiratory suppression or bradycardia in mice. **a, b**, Muzepan1 produces respiratory suppression (**a**) and bradycardia (**b**) at intraperitoneal doses of 24 mg kg⁻¹ and 48 mg kg⁻¹ but not at 3 mg kg⁻¹ in male and female mice. Neither dose alters fentanyl (0.3 mg kg⁻¹ or 2 mg kg⁻¹, intraperitoneal) effects when combined. **c, d**, The AUC for the 1.5 h after drug treatments (doses in mg kg⁻¹, intraperitoneal,

indicated in brackets) for oxygen saturation (**c**) and heart rate (**d**). One-way ANOVA with Tukey's post hoc test. See Extended Data Table 5 for two-way repeated measures ANOVA for time course data in **a, b** and number and sex of mice in each assay ($n = 4-8$; individual mouse data are shown as symbols in the bar charts).

(RGS) family proteins. However, if the receptor can promote GTP binding and release independent of the energy expenditure of hydrolysis, it may present a means to affect G-protein coupling kinetics which would change the effect of such GAPs in the system.

Our findings show that the receptor can facilitate GTP release; however, the context of the environment and the nature of the ligand will determine the ultimate consolidation of signalling. Therefore, the bidirectionality of the exchange function of a GPCR will affect the overall kinetics and energy landscape of GPCR activation. Thus, we propose a modification of the two-state model of receptor activation (as we present in Fig. 1a). We believe the net effect of an agonist on the fate of the G protein is a sum of the hydrolysis and the exchange, and not simply the forward reaction limited to energy-consuming hydrolysis.

The nature of the GTPγS binding assay favours the detection of coupling to inhibitory G proteins owing to their abundance in cells; moreover, the actions at the MOR can be blocked by pertussis toxin, further implicating the Gα_{i/o} class of proteins¹⁴. It remains to be determined whether this occurs for other G proteins (such as, G_s or G_q proteins). In addition to having a role in determining preference for G-protein signalling, the release cycle of one G protein could shift the preference of the receptor away from another G protein. Effectively, the release cycle of a previously coupled G protein would also compete with the

coupling of another class of G protein (that is, the inability to change coupling from G_i to G_q). One benefit of this approach is that it allows for the simultaneous assessment of two active states of the receptor in a single receptor population and that it can be amenable to testing in tissue. In the spinal cord, we note that DAMGO, an MOR-selective enkephalin analogue, is more potent in promoting GTP binding than release, whereas in the CHO overexpression cell lines, the DAMGO effects are nearly equivalent in both assays. Other GPCRs examined in the CHO cell lines may also prove to have differential affinities for one active state over the other when tested in the endogenous setting.

Our studies have shown that compounds that prefer the release state of the MOR are able to enhance and prolong the effects of opioid analgesics in mouse thermal nociception assays. We favour a model in which the receptor engages in multiple exchange events, keeping the G protein nearby to rapidly exchange nucleotide, and allowing another cycle to begin limiting the expenditure of energy through GTPase activity. This would enable the receptor-mediated G-protein activation to be reset without waiting for the completion of the GTPase cycle and, further, it keeps the proteins in close proximity. Of note, SR-17018 is also release-preferring, as well as a partial agonist, and a G-protein-signalling-biased agonist (over β-arrestin2 recruitment). In mice, it produces antinociception without respiratory suppression;

moreover, it does not lead to tolerance in several pain assays^{24,29,30}. In addition to being a G-protein-signalling-biased partial agonist, SR-17018 has also been shown to be a noncompetitive agonist that stabilizes G-protein signalling¹⁶. Moreover, treatment of morphine-tolerant mice with SR-17018 leads to a reversal of morphine tolerance while suppressing signs of withdrawal and restoring morphine sensitivity in mice²⁹. It is not yet clear what exact properties lead to SR-17018 restoration of sensitivity but the identification of this property and the further generation of probe compounds that preserve or eliminate each property independently will aid our studies to disentangle desirable opioid effects from unwanted side effects.

Notably, muzepan1 produces respiratory suppression and bradycardia at 24 and 48 mg kg⁻¹, which may not be unexpected, as it is a full agonist that does not show a preference for G-protein signalling over β -arrestin2 recruitment. However, whereas it is not surprising that the inactive dose of 3 mg kg⁻¹ muzepan1 does not produce an additive effect with 0.3 or 2 mg kg⁻¹ fentanyl, it is surprising that 24 mg kg⁻¹ dose of muzepan1, which alone does suppress activity, does not enhance or prolong the effects of 2 mg kg⁻¹ fentanyl on respiratory suppression or bradycardia. There remains the potential that we have reached a ceiling effect with the mice when combining the higher doses of each drug. However, we do note that the median lethal dose (LD₅₀) for fentanyl has been recorded at 113 mg kg⁻¹ (intraperitoneal) in C57BL/6 mice, suggesting that there should be potential to detect an additive decrease in oxygen saturation and heart rate³¹. Notably, the muzepons have been explored here for their robust ability to produce GTP release, and they may not be considered safer opioids at this time, as we clearly show that significant respiratory suppression can be observed with muzepan1 at high doses. Moreover, these probe compounds have not been evaluated for other opioid side effects, such as addiction liability, tolerance or dependence, nor have they been evaluated for general safety or toxicity. However, developing agonists that preserve the release preference, while also implementing other favourable properties, such as slow onset pharmacokinetics, partial agonism and fine-tuning preference for different signalling partners (β -arrestin versus G protein) is underway to improve the safety profile of such ligands.

With the examples provided here, we can conclude that at nearly inactive doses, muzepan1 enhances fentanyl-induced antinociception but does not enhance fentanyl-induced respiratory suppression in mice. We speculate that by favouring a release state of the receptor, either the receptor or the G proteins may have less opportunity to engage with secondary effectors that promote the cardiovascular side effects while perpetuating signalling that leads to antinociception. For example, a G protein that is cycling between GTP-bound and GDP-bound states may have more or fewer opportunities to interact with the G $\beta\gamma$ subunits, thereby preventing their further interaction with ion channels or other effectors, such as regulatory kinases or scaffolding proteins. Alternatively, the perpetual engagement between the receptor and G protein could sterically prevent engagement with other proteins, such as β -arrestins or other G-protein types. In this manner, one downstream signalling pathway may become more efficient while another is disengaged. In the case of SR-17018, a preference for promoting G-protein binding over β -arrestin2 recruitment as well as a preference over GIRK activation have been demonstrated^{24,32,33}. It remains to be determined what preferences will be desirable; however it is evident that the environment of the receptor in the tissue in which it mediates its response will dictate any preference for a secondary cascade.

By demonstrating that there are changes in both the rank order potency and efficacy of agonists, we demonstrate that there is a change in activity of the receptor for the GTP binding versus GTP release functions. Further, we conclude that the GTP binding and release events are independently regulated by receptor activity and that an agonist has the opportunity to not only promote G-protein activation but also to modulate G-protein activity in a state-selective manner. We demonstrate that a GTP-releasing agonist can enhance and prolong the

effects of conventional opioid agonists in antinociception without enhancing respiratory suppression or bradycardia in mice. Therefore, the selective modulation of potency and efficacy in the release function can change the dynamics of the drug effects in vivo. More broadly, since a ligand can induce a preference for exchange selectivity at a particular receptor, the agonist could thereby effectively regulate the availability of the G α subunit for subsequent protein interactions and downstream signalling events. Overall, these findings demonstrate that for the opioid receptors, and possibly for all GPCRs, drug action will be a composite of its ability to promote both GTP binding and GTP release from the G protein.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-025-09880-5>.

- Helmreich, E. J. & Hofmann, K. P. Structure and function of proteins in G-protein-coupled signal transfer. *Biochim. Biophys. Acta* **1286**, 285–322 (1996).
- Cassel, D. & Selinger, Z. Catecholamine-induced release of [³H]-Gpp(NH)p from turkey erythrocyte adenylate cyclase. *J. Cyclic Nucleotide Res.* **3**, 11–22 (1977).
- Cassel, D., Levkovitz, H. & Selinger, Z. The regulatory GTPase cycle of turkey erythrocyte adenylate cyclase. *J. Cyclic Nucleotide Res.* **3**, 393–406 (1977).
- Rodbell, M., Birnbaumer, L., Pohl, S. L. & Krans, H. M. The glucagon-sensitive adenyl cyclase system in plasma membranes of rat liver. V. An obligatory role of guanylnucleotides in glucagon action. *J. Biol. Chem.* **246**, 1877–1882 (1971).
- Ross, E. M. & Gilman, A. G. Reconstitution of catecholamine-sensitive adenylate cyclase activity: interactions of solubilized components with receptor-replete membranes. *Proc. Natl Acad. Sci. USA* **74**, 3715–3719 (1977).
- Hildebrandt, J. D., Codina, J., Risinger, R. & Birnbaumer, L. Identification of a γ subunit associated with the adenylate cyclase regulatory proteins N₁ and N₂. *J. Biol. Chem.* **259**, 2039–2042 (1984).
- Limbird, L. E., Gill, D. M. & Lefkowitz, R. J. Agonist-promoted coupling of the beta-adrenergic receptor with the guanine nucleotide regulatory protein of the adenylate cyclase system. *Proc. Natl Acad. Sci. USA* **77**, 775–779 (1980).
- Brandt, D. R. & Ross, E. M. Catecholamine-stimulated GTPase cycle. Multiple sites of regulation by beta-adrenergic receptor and Mg²⁺ studied in reconstituted receptor-G_s vesicles. *J. Biol. Chem.* **261**, 1656–1664 (1986).
- Gilman, A. G. G proteins: transducers of receptor-generated signals. *Annu. Rev. Biochem.* **56**, 615–649 (1987).
- Tolkovsky, A. M. & Levitzki, A. Coupling of a single adenylate cyclase to two receptors: adenosine and catecholamine. *Biochemistry* **17**, 3811–3817 (1978).
- Bennett, N. & Dupont, Y. The G-protein of retinal rod outer segments (transducin). Mechanism of interaction with rhodopsin and nucleotides. *J. Biol. Chem.* **260**, 4156–4168 (1985).
- May, D. C. & Ross, E. M. Rapid binding of guanosine 5'-O-(3-thiotriphosphate) to an apparent complex of beta-adrenergic receptor and the GTP-binding regulatory protein G_s. *Biochemistry* **27**, 4888–4893 (1988).
- Traynor, J. R., Clark, M. J. & Remmers, A. E. Relationship between rate and extent of G protein activation: comparison between full and partial opioid agonists. *J. Pharmacol. Exp. Ther.* **300**, 157–161 (2002).
- Bohn, L. M. & Stahl, E. L. Characterization of the GTP γ S release function of a G protein-coupled receptor. *Nat. Commun.* <https://doi.org/10.1038/s41467-025-66516-y> (2025).
- Asano, T. & Ross, E. M. Catecholamine-stimulated guanosine 5'-O-(3-thiotriphosphate) binding to the stimulatory GTP-binding protein of adenylate cyclase: kinetic analysis in reconstituted phospholipid vesicles. *Biochemistry* **23**, 5467–5471 (1984).
- Stahl, E. L., et al. G protein signaling-biased mu opioid receptor agonists that produce sustained G protein activation are noncompetitive agonists. *Proc. Natl Acad. Sci. USA* **118**, e2102178118 (2021).
- Kenakin, T. & Christopoulos, A. Signalling bias in new drug discovery: detection, quantification and therapeutic impact. *Nat. Rev. Drug Discov.* **12**, 205–216 (2013).
- DeWire, S. M. et al. A G protein-biased ligand at the mu-opioid receptor is potentially analgesic with reduced gastrointestinal and respiratory dysfunction compared with morphine. *J. Pharmacol. Exp. Ther.* **344**, 708–717 (2013).
- Manglik, A. et al. Structure-based discovery of opioid analgesics with reduced side effects. *Nature* **537**, 185–190 (2016).
- Groer, C. E. et al. An opioid agonist that does not induce mu-opioid receptor-arrestin interactions or receptor internalization. *Mol. Pharmacol.* **71**, 549–557 (2007).
- Ehrlich, A. T. et al. Biased signaling of the mu opioid receptor revealed in native neurons. *iScience* **14**, 47–57 (2019).
- McPherson, J. et al. Mu-opioid receptors: correlation of agonist efficacy for signalling with ability to activate internalization. *Mol. Pharmacol.* **78**, 756–766 (2010).
- Pedersen, M. F. et al. Biased agonism of clinically approved mu-opioid receptor agonists and TRV130 is not controlled by binding and signaling kinetics. *Neuropharmacology* **166**, 107718 (2020).
- Schmid, C. L. et al. Bias factor and therapeutic window correlate to predict safer opioid analgesics. *Cell* **171**, 1165–1175.e1113 (2017).